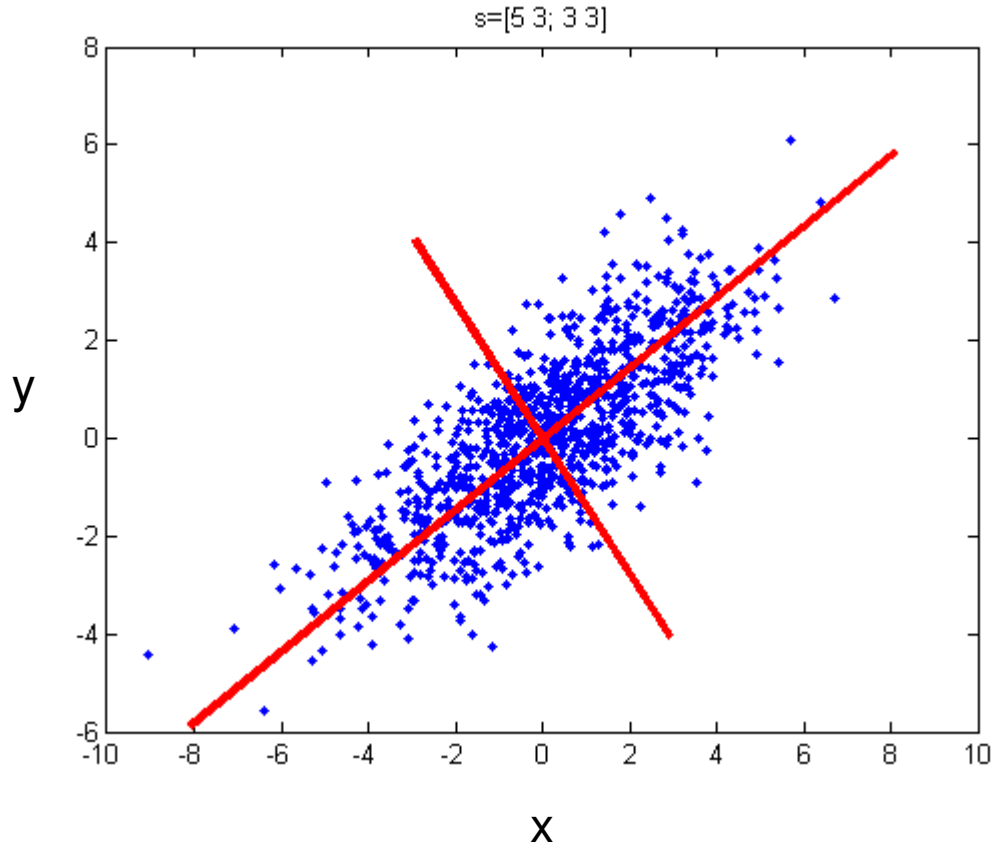
Component Analysis and Discriminants

Component Analysis & Discriminants

- One approach to coping with the problem of excessive dimensionality is to reduce the dimensionality by combining features.
- Linear combinations are particularly attractive because they are simple to compute and analytically tractable. In effect, linear methods project the high-dimensional data onto a lower dimensional space.
- There are two classical approaches to finding effective linear transformations.
  - Principal Component Analysis (PCA): a projection that best represents the data in the least-squares sense.
  - Multiple Discriminant Analysis (MDA): a projection that best separates the data in the least squares sense.



$\boldsymbol{\mu} = [2, 1]$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix}$$

Eigenvectors: $\begin{bmatrix} 0.5863 & -0.8101 \\ -0.8101 & -0.5863 \end{bmatrix}$

Eigenvalues: $\begin{bmatrix} 0.8344 & 0 \\ 0 & 6.9753 \end{bmatrix}$

Principal Component Analysis (PCA)

- Consider the problem of representing the *n* *d*-dim. samples $\mathbf{x_1},\ldots,\mathbf{x_n}$ by a single vector $\mathbf{x_0}$.
- Find $\mathbf{x_0}$ such that the sum of the squared distances between $\mathbf{x_0}$ and all $\mathbf{x_k}$ is minimized
- Define the squared-error criterion function $J_0(\mathbf{x_0})$ by

$$J_o\left(\mathbf{x_0}\right)=\sum_{k=1}^{n}\left\|\mathbf{x_0}-\mathbf{x_k}\right\|^2,$$

  and seek the value of $\mathbf{x_0}$ that minimizes J_0.
- The solution to this problem is given by $\mathbf{x_0}=\mathbf{m}$, where $\mathbf{m}$ is the sample mean,

$$\mathbf{m}=\frac{1}{n}\sum_{k=1}^{n}\mathbf{x_k}.$$

Principal Component Analysis

This can be easily verified by writing

$$J_0(\mathbf{x_0}) = \sum_{k=1}^{n} \| (\mathbf{x_0} - \mathbf{m}) - (\mathbf{x_k} - \mathbf{m}) \|^2$$

$$= \sum_{k=1}^{n} \| (\mathbf{x_0} - \mathbf{m}) \|^2 - 2\sum_{k=1}^{n} (\mathbf{x_0} - \mathbf{m})^t (\mathbf{x_k} - \mathbf{m}) + \sum_{k=1}^{n} \| (\mathbf{x_k} - \mathbf{m}) \|^2$$

$$= \sum_{k=1}^{n} \| (\mathbf{x_0} - \mathbf{m}) \|^2 - 2(\mathbf{x_0} - \mathbf{m})^t \sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{m}) + \sum_{k=1}^{n} \| (\mathbf{x_k} - \mathbf{m}) \|^2$$

$$= \sum_{k=1}^{n} \| (\mathbf{x_0} - \mathbf{m}) \|^2 + \underbrace{\sum_{k=1}^{n} \| (\mathbf{x_k} - \mathbf{m}) \|^2}_{\text{Independent of } \mathbf{x_0}}.$$

Since the second sum is independent of $\mathbf{x_0}$, this expression is minimized by the choice $\mathbf{x_0} = \mathbf{m}$.

Principal Component Analysis

The sample mean is a zero-dimensional representation of the data set. While it is simple, it does not reveal any of the variability in the data.

To obtain a more interesting one-dimensional representation, project the data onto a line running through the sample mean. Let e be a unit vector in the direction of the line. Then equation of the line can be written as

$$\mathbf{x} = \mathbf{m} + a\mathbf{e},$$

where the scalar a corresponds to the distance of any point x from the mean **m**. If we represent $\mathbf{x_k}$ by $\mathbf{m}+a_k\mathbf{e}$, we can find an "optional" set of coefficients a_k by minimizing the squared-error criterion function

$$J_1(a_1,...,a_n,\mathbf{e}) = \sum_{k=1}^{n} \| (\mathbf{m}+a_k\mathbf{e}) - \mathbf{x_k} \|^2 = \sum_{k=1}^{n} \| a_k\mathbf{e} - (\mathbf{x_k} - \mathbf{m}) \|^2$$

$$= \sum_{k=1}^{n} a_k^2 \| \mathbf{e} \|^2 - 2\sum_{k=1}^{n} a_k \mathbf{e}^t (\mathbf{x_k} - \mathbf{m}) + \sum_{k=1}^{n} \|(\mathbf{x_k} - \mathbf{m}) \|^2 .$$

Principal Component Analysis

- Recognizing that $\|\mathbf{e}\| = 1$, partially differentiating with respect to a_k, and setting the derivative to zero, we obtain

$$a_k = \mathbf{e}^t (\mathbf{x_k} - \mathbf{m}).$$

- Geometrically, this result says that to obtain a least-squares solution, project the vectors $\mathbf{x_k}$ onto the line in the direction of e that passes through the sample mean.

- What is the *best direction* e for the line? The solution involves the *scatter matrix* **S**

$$\mathbf{S} = \sum_{k=1}^{n} (\mathbf{x_k} - \mathbf{m})(\mathbf{x_k} - \mathbf{m})^t.$$

Principal Component Analysis

- The scatter matrix is merely (n-1) times the sample covariance matrix. It arises here when we substitute a_k found in Eq. (83) into Eq. (82) to obtain

$$
\begin{aligned}
J_1(\mathbf{e}) &= \sum_{k=1}^{n} a_k^2 - 2\sum_{k=1}^{n} a_k^2 + \sum_{k=1}^{n} \| \mathbf{x_k} - \mathbf{m} \|^2 \\
&= -\sum_{k=1}^{n} [\mathbf{e}^t (\mathbf{x_k} - \mathbf{m})]^2 + \sum_{k=1}^{n} \| \mathbf{x_k} - \mathbf{m} \|^2 \\
&= -\sum_{k=1}^{n} \mathbf{e}^t (\mathbf{x_k} - \mathbf{m})(\mathbf{x_k} - \mathbf{m})^t \mathbf{e} + \sum_{k=1}^{n} \| \mathbf{x_k} - \mathbf{m} \|^2 \\
&= -\mathbf{e}^t \mathbf{S} \mathbf{e} + \sum_{k=1}^{n} \| \mathbf{x_k} - \mathbf{m} \|^2 .
\end{aligned}
$$

Principal Component Analysis

- The vector e that minimizes J_1 also maximizes $\mathbf{e}^t\mathbf{S}\mathbf{e}$. We use the method of Lagrange multipliers (Section A.3 of the Appendix) to maximize $\mathbf{e}^t\mathbf{S}\mathbf{e}$ subject to the constraint that $\|\mathbf{e}\| = 1$. Letting λ be the undetermined multiplier, we differentiate

$$u = \mathbf{e}^t\mathbf{S}\mathbf{e} - \lambda(\mathbf{e}^t\mathbf{e} - 1)$$

  with respect to **e** to obtain

$$\frac{\partial u}{\partial \mathbf{e}} = 2\mathbf{S}\mathbf{e} - 2\lambda\mathbf{e}.$$

- Setting this gradient vector equal to zero, **e** is the eigenvector of the scatter matrix:

$$\mathbf{S}\mathbf{e} = \lambda\mathbf{e}.$$

Principal Component Analysis

- Since $\mathbf{e}^t\mathbf{S}\mathbf{e} = \lambda\, \mathbf{e}^t\mathbf{e} = \lambda$, it follows that to maximize $\mathbf{e}^t\mathbf{S}\mathbf{e}$, we want to select the eigenvector corresponding to the largest eigenvalue of the scatter matrix.
- In other words, to find the best one-dimensional projection of the *d*-dimensional data (in the least-sum-of-squared-error sense), project the data onto a line through the sample mean in the direction of the eigenvector of the scatter matrix with the largest eigenvalue.

Principal Component Analysis

- This result can be readily extended from a one-dimensional projection to a d'-dimensional projection ($d'<d$). In place of Eq. (81), we write

$$\mathbf{x} = \mathbf{m} + \sum_{i=1}^{d'} a_i \mathbf{e_i},$$

where $d' \leq d$.

- It is not difficult to show that the criterion function

$$J_{d'} = \sum_{k=1}^{n} \left\| (\mathbf{m} + \sum_{i=1}^{d'} a_{ki} \mathbf{e_i}) - \mathbf{x_k} \right\|^2$$

is minimized when the vectors $\mathbf{e_1}, \ldots \mathbf{e_{d'}}$ are the d' eigenvectors of $\mathbf{S}$ with the largest eigenvalues.

Principal Component Analysis

- Because the scatter matrix is real and symmetric, it's eigenvectors are orthogonal. They form a natural set of basis vectors for representing any vector **x**.
- The coefficients a_i in Eq. (89) are the components of x in that basis, called the *principal components*.
- Data points $\mathbf{x_1},\ldots\mathbf{x_n}$ can be viewed as a *d*-dim. hyperellipsoidally shaped cloud; eigenvectors of the scatter matrix are the principal axes of hyperellipsoid.
- PCA reduces the feature dimensionality by restricting attention to those directions along which the scatter of the cloud is greatest.

Eigen Decomposition

- Given a linear transformation **A**, a non-zero vector *w* is an eigen-vector of **A** if it satisfies the *eigenvalue* equation for some scalar λ.

$$\mathbf{A}\mathbf{w} = \lambda\mathbf{w}$$

Solution:

$$\begin{aligned}
&\mathbf{A}\mathbf{w} - \lambda\mathbf{I}\mathbf{w} = 0 \\
&\Rightarrow (\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = 0 \\
&\Rightarrow \det(\mathbf{A} - \lambda\mathbf{I}) = 0
\end{aligned}$$

(Characteristic equation)

Ex 1. $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

Eigenvalue:

$$\begin{aligned}
&\det\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = 0 \\
&(2-\lambda)^2 - 1 = 0 \\
&\lambda^2 - 4\lambda + 3 = 0 \\
&\Rightarrow \lambda_1 = 1 \text{ and } \lambda_2 = 3
\end{aligned}$$

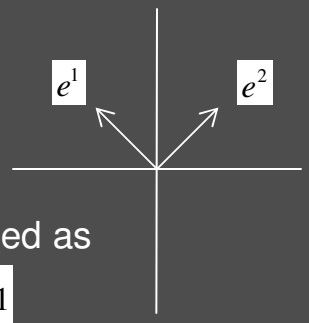
Eigenvector:

$$\begin{bmatrix} 2-\lambda_1 & 1 \\ 1 & 2-\lambda_1 \end{bmatrix}\begin{bmatrix} e_1^1 \\ e_2^1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2-\lambda_2 & 1 \\ 1 & 2-\lambda_2 \end{bmatrix}\begin{bmatrix} e_1^2 \\ e_2^2 \end{bmatrix} = 0$$

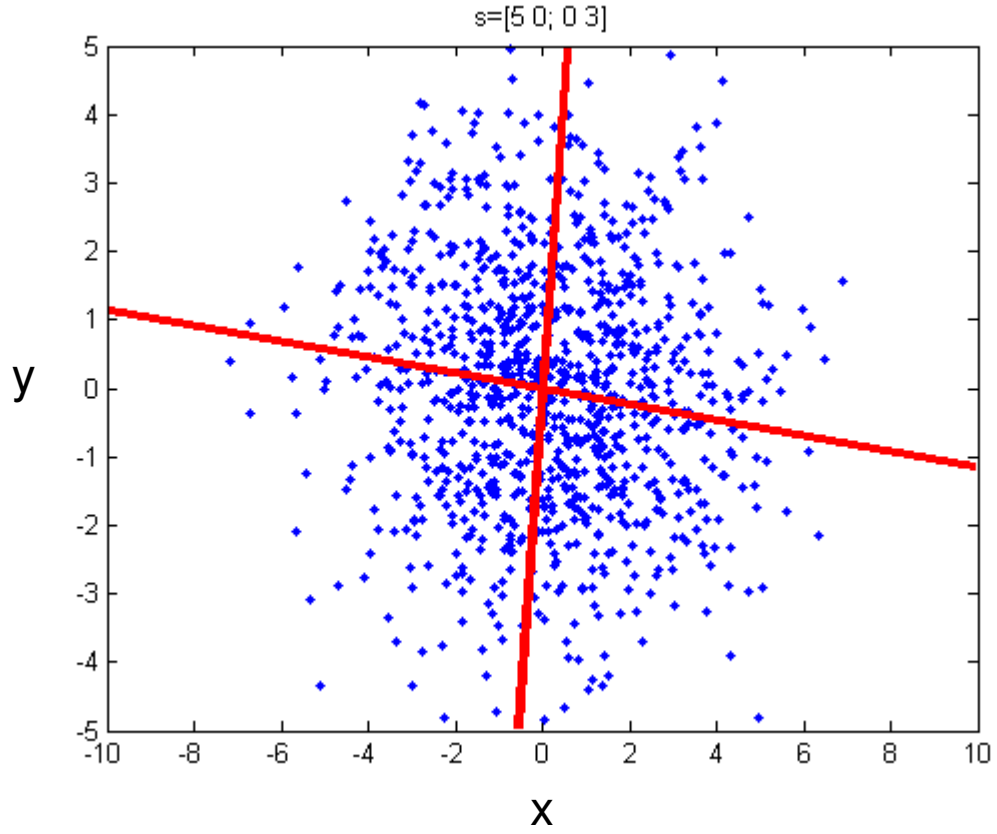
$$\begin{bmatrix} e_1^1 \\ e_2^1 \end{bmatrix} = \begin{bmatrix} -0.7071 \\ 0.7071 \end{bmatrix}$$

$$\begin{bmatrix} e_1^2 \\ e_2^2 \end{bmatrix} = \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}$$



Eigenvector is normalized as

$$\sqrt{e_1^2 + e_2^2} = 1$$

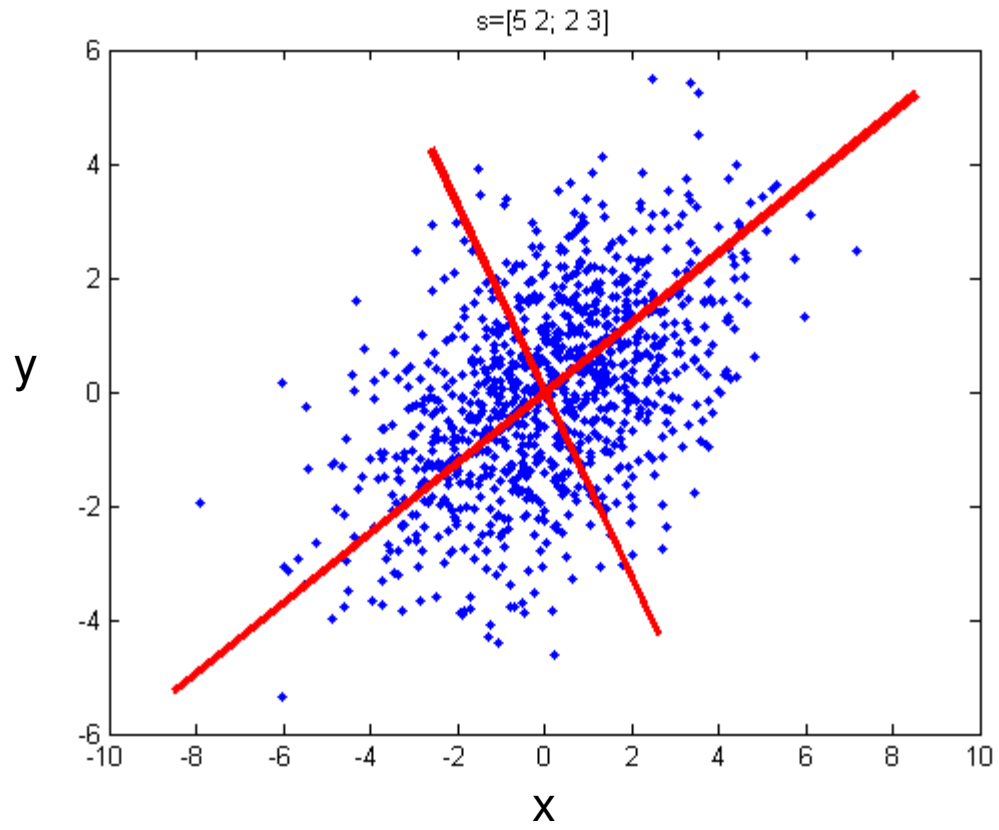


$\boldsymbol{\mu} = [2, 1]$

$\boldsymbol{\Sigma} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$

Eigenvectors: $\begin{bmatrix} -0.1137 & -0.9935 \\ -0.9935 & 0.1137 \end{bmatrix}$

Eigenvalues: $\begin{bmatrix} 3.1757 & 0 \\ 0 & 5.3882 \end{bmatrix}$

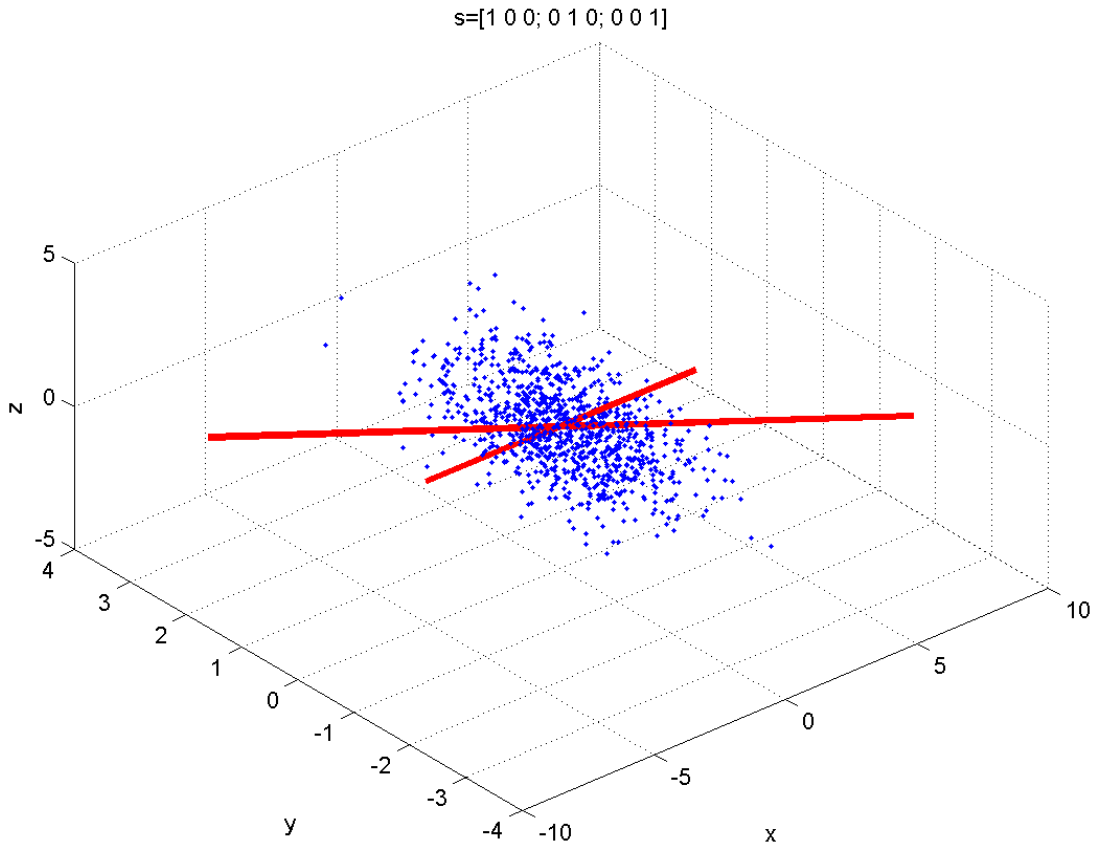


$\boldsymbol{\mu} = [2, 1]$

$\boldsymbol{\Sigma} = \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$

Eigenvectors: $\begin{bmatrix} 0.5238 & -0.8519 \\ -0.8519 & -0.5238 \end{bmatrix}$

Eigenvalues: $\begin{bmatrix} 1.7230 & 0 \\ 0 & 5.6644 \end{bmatrix}$



$\boldsymbol{\mu} = [4, 2, 1]$

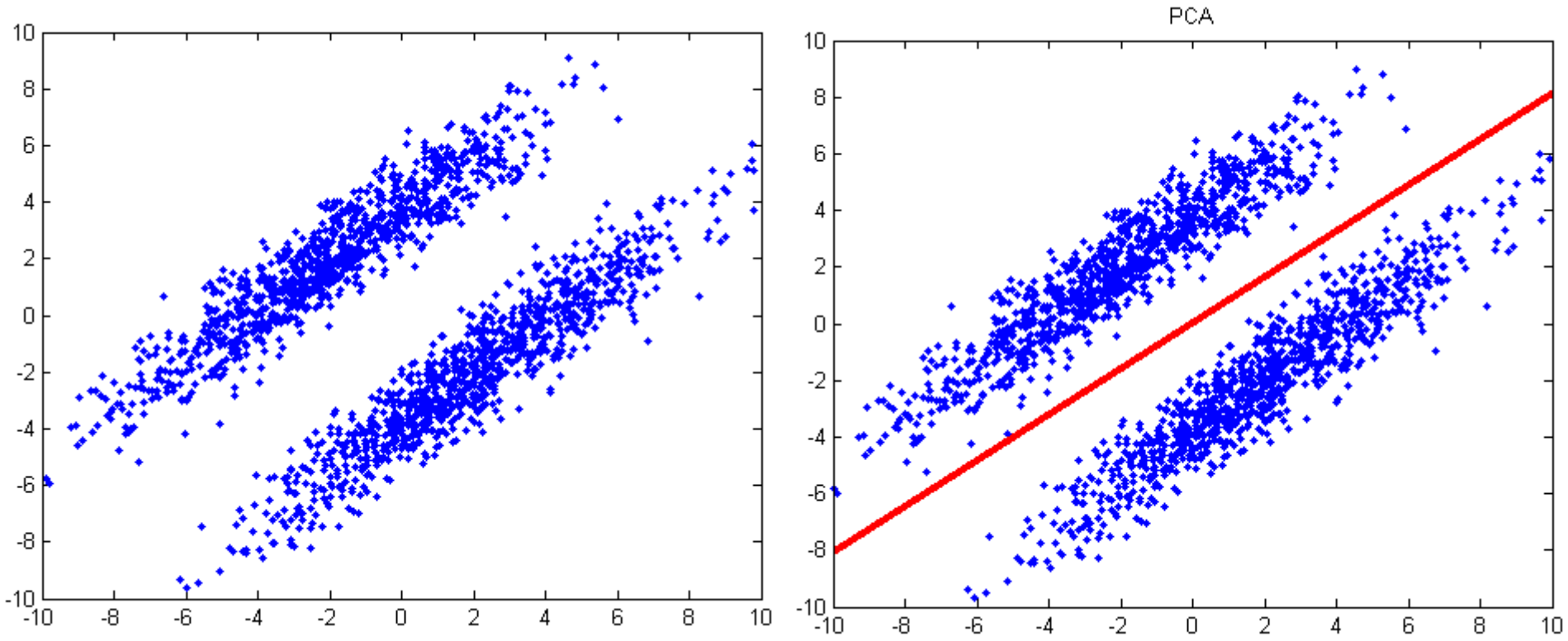
$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Eigenvectors: $\begin{bmatrix} 0.2190 & 0.0522 & -0.9743 \\ 0.8735 & -0.4554 & 0.1720 \\ 0.4347 & 0.8888 & 0.1453 \end{bmatrix}$

Eigenvalues: $\begin{bmatrix} 480.4256 & 0 & 0 \\ 0 & 498.6763 & 0 \\ 0 & 0 & 568.5106 \end{bmatrix}$

PCA

$$Y = w^T X$$



Find a transformation **w**, such that the $\mathbf{w}^T\mathbf{x}$ is dispersed the most (maximum distribution)

Face Representation using PCA/LDA

Input face



EigenFaces



56.4 38.6 -19.7 9.8 -45.9 19.6 - 14.2 . . .

Fisherfaces



18.3 35.6 -17.5 -27.6 60.6 -20.8 41.9 -9.6 . . .

PCA LDA

Reconstructed face





Minimize reconstruction error

Maximize between-class to within-class scatter

Fisher Linear Discriminant

- PCA finds components that are useful for explaining the variance in the data, but these components may not be useful for discrimination between different classes.
- Since we pool all the samples in computing PCA (no category label), components discarded by PCA might be exactly those that are needed for distinguishing between classes.
- Whereas PCA seeks directions that are efficient for representation, *discriminant analysis* seeks directions that are efficient for discrimination.

Fisher Linear Discriminant

- Consider the problem of projecting labeled data from *d* dimensions onto a line.
- Even if the samples formed well-separated, compact clusters in *d*-space, projection onto an arbitrary line will usually produce a confused mixture of samples from all of the classes, leading to poor classification.
- However, by moving the projection line around, we might be able to find an orientation for which the projected samples are well separated. This is exactly the goal of discriminant analysis.

Fisher Linear Discriminant

- Given *n d*-dimensional samples $\mathbf{x_1}$, ..., $\mathbf{x_n}$; n_1 in the subset D_1 labeled ω_1 and n_2 in the subset D_2 labeled ω_2. If we form a linear combination of the components of **x**, we obtain a corresponding set of n samples y_1, ..., y_n divided into the subsets Y_1 and Y_2.

$$y = \mathbf{w}^t \mathbf{x}$$

- Geometrically, if ||**w**||= 1, each y_i is the projection of the corresponding $\mathbf{x_i}$ onto a line in the direction of **w**. Note, the magnitude of **w** is of no real significance, since it merely scales *y*. The direction of **w** is important, however.
- If we imagine that the samples labeled ω_1 fall more or less into one cluster while those labeled ω_2 fall in another in *d* dimensions, we want the projections falling onto the line to be well separated as well.

Fisher Linear Discriminant

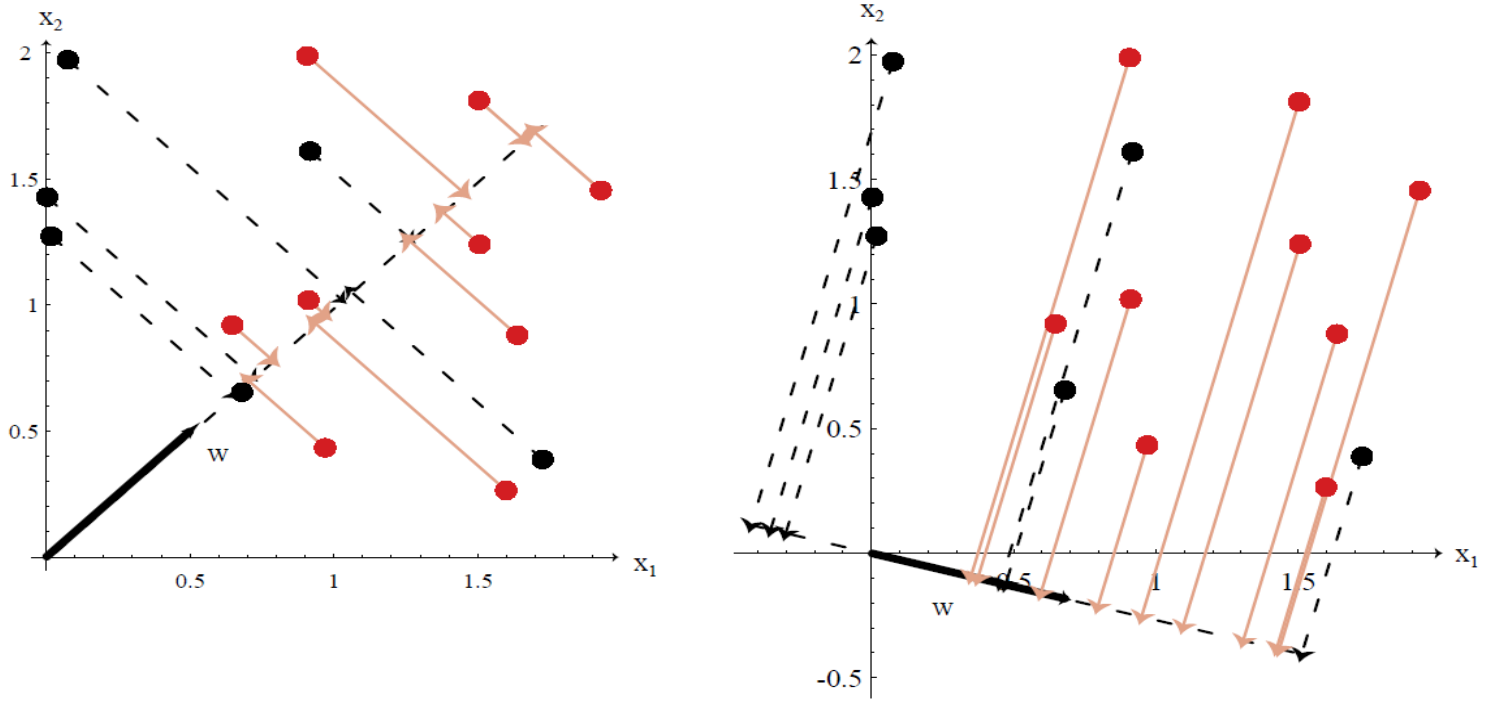


Figure 3.5 illustrates the effect of choosing two different values for **w** for a two-dimensional example. If the original distributions are multimodal and highly overlapping, even the "best" **w** is unlikely to provide adequate separation.

Fisher Linear Discriminant

- How to find the best direction **w** that will enable accurate classification?
- A measure of the separation between the projected points is the difference of the sample means. If $\mathbf{m_i}$ is the *d*-dimensional sample mean

$$\mathbf{m_i} = \frac{1}{n_i} \sum_{\mathbf{x} \in D_i} \mathbf{x},$$

then the sample mean for the projected points is

$$\widetilde{m}_i = \frac{1}{n_i} \sum_{y \in Y_i} y$$

$$= \frac{1}{n_i} \sum_{x \in D_i} \mathbf{w}^t \mathbf{x} = \mathbf{w}^t \mathbf{m_i}$$

and is simply the projection of $\mathbf{m_i}$.

Fisher Linear Discriminant

- The distance between the projected means is

$$|\tilde{m}_1 - \tilde{m}_2| = |\mathbf{w}^t(\mathbf{m}_1 - \mathbf{m}_2)|,$$

  and we can make this difference as large as we wish merely by scaling **w**.
- To obtain good separation of the projected data we really want the difference between the means to be large relative to some measure of the standard deviations for each class.
- Define the *scatter* for projected samples labeled ω_i by

$$\tilde{s}_i^2 = \sum_{y \in \mathcal{Y}_i} (y - \tilde{m}_i)^2.$$

Fisher Linear Discriminant

- Thus, $(1/n)(\tilde{s}_1^2 + \tilde{s}_2^2)$ is an estimate of the variance of the pooled data, and $\tilde{s}_1^2 + \tilde{s}_2^2$ is called the total *within-class scatter* of the projected samples. The *Fisher linear discriminant* employs that linear function $\mathbf{w}^t\mathbf{x}$ for which the criterion function

$$J(\mathbf{w}) = \frac{|\tilde{m}_1 - \tilde{m}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

  is maximum (and independent of $||\mathbf{w}||$).

- The vector w maximizing $J(\cdot)$ leads to the best separation between the two projected sets.

- How to solve for the optimal **w**?

Fisher Linear Discriminant

To obtain $J(\cdot)$ as an explicit function of $\mathbf{w}$, we define the scatter matrices $\mathbf{S_i}$ and scatter $\mathbf{S_W}$ by

$$\mathbf{S}_i = \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^t$$

and

$$\mathbf{S}_W = \mathbf{S}_1 + \mathbf{S}_2.$$

Then we can write

$$\begin{aligned} \tilde{s}_i^2 &= \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{w}^t \mathbf{x} - \mathbf{w}^t \mathbf{m}_i)^2 \\ &= \sum_{\mathbf{x} \in \mathcal{D}_i} \mathbf{w}^t (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^t \mathbf{w} \\ &= \mathbf{w}^t \mathbf{S}_i \mathbf{w}; \end{aligned}$$

Fisher Linear Discriminant

The sum of these scatters can be written

$$\tilde{s}_1^2 + \tilde{s}_2^2 = \mathbf{w}^t \mathbf{S}_W \mathbf{w}.$$

Similarly, the separations of the projected means obeys

$$\begin{aligned}(\tilde{m}_1 - \tilde{m}_2)^2 &= (\mathbf{w}^t \mathbf{m}_1 - \mathbf{w}^t \mathbf{m}_2)^2 \\ &= \mathbf{w}^t (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^t \mathbf{w} \\ &= \mathbf{w}^t \mathbf{S}_B \mathbf{w},\end{aligned}$$

where

$$\mathbf{S}_B = (\mathbf{m}_1 - \mathbf{m}_2)(\mathbf{m}_1 - \mathbf{m}_2)^t.$$

Fisher Linear Discriminant

- $\mathbf{S_W}$ is called the *within-class scatter matrix*. It is proportional to the sample covariance matrix for the pooled *d*-dimensional data. It is symmetric and positive semidefinite, and is usually nonsingular if $n > d$.
- $\mathbf{S_B}$ is called the *between-class scatter matrix*. It is also symmetric and positive semidefinite, but because it is the outer product of two vectors, its rank is at most one.
- In particular for any $\mathbf{w}$, $\mathbf{S_B w}$ is in the direction of $(\mathbf{m_1} - \mathbf{m_2})$.

Fisher Linear Discriminant

- In terms of $\mathbf{S_B}$ and $\mathbf{S_W}$, the criterion function $J(\cdot)$ can be written as

$$J(\mathbf{w}) = \frac{\mathbf{w}^t \mathbf{S}_B \mathbf{w}}{\mathbf{w}^t \mathbf{S}_W \mathbf{w}}.$$

- It is easy to show that a vector **w** that maximizes $J(\cdot)$ must satisfy

$$\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w},$$

  for some constant λ, which is a generalized eigenvalue problem.

- If $\mathbf{S_W}$ is nonsingular we can obtain a conventional eigenvalue problem by writing

$$\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w}.$$

Fisher Linear Discriminant

In our particular case, it is unnecessary to solve for the eigenvalues and eigenvectors of $\mathbf{S}_W^{-1}\mathbf{S}_B$ due to the fact that $\mathbf{S_B w}$ is always in the direction of $\mathbf{m_1 - m_2}$. Since the scale factor for **w** is immaterial, we can immediately write the solution for the w that optimizes $J(\cdot)$:

$$\mathbf{w} = \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2).$$

Fisher Linear Discriminant

- Fisher linear discriminant is the linear function yielding the maximum ratio of between-class scatter to within-class scatter.
- The classification has been converted from a *d*-dimensional problem to a one-dimensional one.
- This mapping is many-to-one, and in theory can not possibly reduce the minimum achievable error rate if we have a very large training set.
- In general, one is willing to sacrifice some of the theoretically attainable performance for the advantages of working in one dimension.
- Find the threshold, i.e., the point along the one-dim. subspace separating the projected points.

Fisher Linear Discriminant

- When the conditional densities $p(\mathbf{x}|\omega_i)$ are multivariate normal with equal covariance $\mathbf{\Sigma}$, the threshold can be computed directly from the optimal decision boundary (Chapter 2)

$$\mathbf{w}^t\mathbf{x} + w_0 = 0$$
$$\mathbf{w} = \mathbf{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2),$$

  where w_0 is a constant involving w and the prior.

- Thus, for the normal, equal-covariance case, the optimal decision rule is merely to decide ω_1 if Fisher's linear discriminant exceed some threshold, and to decide ω_2 otherwise.

Multiple Discriminant Analysis

- Generalize 2-class Fisher's linear discriminant to *c*-class problem.
- Now, the projection is from a *d*-dimensional space to a (*c* - 1)-dimensional space, *d* ≥ *c*.
- The generalization for the within-class scatter matrix is obvious:

$$\mathbf{S}_W = \sum_{i=1}^{c} \mathbf{S}_i$$

where, as before,

$$\mathbf{S}_i = \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^t$$

and

$$\mathbf{m}_i = \frac{1}{n_i} \sum_{\mathbf{x} \in \mathcal{D}_i} \mathbf{x}.$$

Multiple Discriminant Analysis

- How to generalize $\mathbf{S_B}$ for *c* classes?
- Define a *total mean vector* m and a *total scatter matrix* $\mathbf{S_T}$ by

$$\mathbf{m}_i = \frac{1}{n_i} \sum_{\mathbf{x} \in \mathcal{D}_i} \mathbf{x}.$$

and

$$\mathbf{m} = \frac{1}{n} \sum_{\mathbf{x}} \mathbf{x} = \frac{1}{n} \sum_{i=1}^{c} n_i \mathbf{m}_i$$

- Then it follows that (see Eq. (113) in the book)

$$\begin{aligned}
\mathbf{S}_T &= \sum_{i=1}^{c} \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{x} - \mathbf{m}_i + \mathbf{m}_i - \mathbf{m})(\mathbf{x} - \mathbf{m}_i + \mathbf{m}_i - \mathbf{m})^t \\
&= \sum_{i=1}^{c} \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^t + \sum_{i=1}^{c} \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^t \\
&= \mathbf{S}_W + \sum_{i=1}^{c} n_i (\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^t.
\end{aligned}$$

Multiple Discriminant Analysis

It is natural to define this second term as a general between-class scatter matrix, so that the total scatter is the sum of the within-class scatter and the between-class scatter:

$$\mathbf{S}_B = \sum_{i=1}^{c} n_i(\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^t$$

and

$$\mathbf{S}_T = \mathbf{S}_W + \mathbf{S}_B.$$

If we check the two-class case, we find that the resulting between-class scatter matrix is n_1n_2/n times our previous definition.$*$

$*$ *We could redefine S_B for the two-class case to obtain complete consistency, but there should be* no misunderstanding of our usage.

Multiple Discriminant Analysis

- The projection from a *d*-dimensional space to a *(c-1)*-dimensional space is accomplished by *c*-1 discriminant functions

$$y_i = \mathbf{w}_i^t \mathbf{x} \qquad i = 1, ..., c-1.$$

- If the y_i are viewed as components of a vector *y* and the weight vectors $\mathbf{w_i}$ are viewed as the columns of a *dx(c – 1)* matrix **W**, then the projection can be written as a single matrix equation

$$\mathbf{y} = \mathbf{W}^t \mathbf{x}.$$

Multiple Discriminant Analysis

- The samples x_1, ..., x_n project to a corresponding set of samples y_1, ..., y_n, which can be described by their own mean vectors and scatter matrices. Thus, if we define

$$\tilde{\mathbf{m}}_i = \frac{1}{n_i} \sum_{\mathbf{y} \in \mathcal{Y}_i} \mathbf{y}$$

$$\tilde{\mathbf{m}} = \frac{1}{n} \sum_{i=1}^{c} n_i \tilde{\mathbf{m}}_i$$

$$\tilde{\mathbf{S}}_W = \sum_{i=1}^{c} \sum_{\mathbf{y} \in \mathcal{Y}_i} (\mathbf{y} - \tilde{\mathbf{m}}_i)(\mathbf{y} - \tilde{\mathbf{m}}_i)^t$$

and

$$\tilde{\mathbf{S}}_B = \sum_{i=1}^{c} n_i (\tilde{\mathbf{m}}_i - \tilde{\mathbf{m}})(\tilde{\mathbf{m}}_i - \tilde{\mathbf{m}})^t,$$

it is a straightforward matter to show that

$$\tilde{\mathbf{S}}_W = \mathbf{W}^t \mathbf{S}_W \mathbf{W}$$

and

$$\tilde{\mathbf{S}}_B = \mathbf{W}^t \mathbf{S}_B \mathbf{W}.$$

Multiple Discriminant Analysis

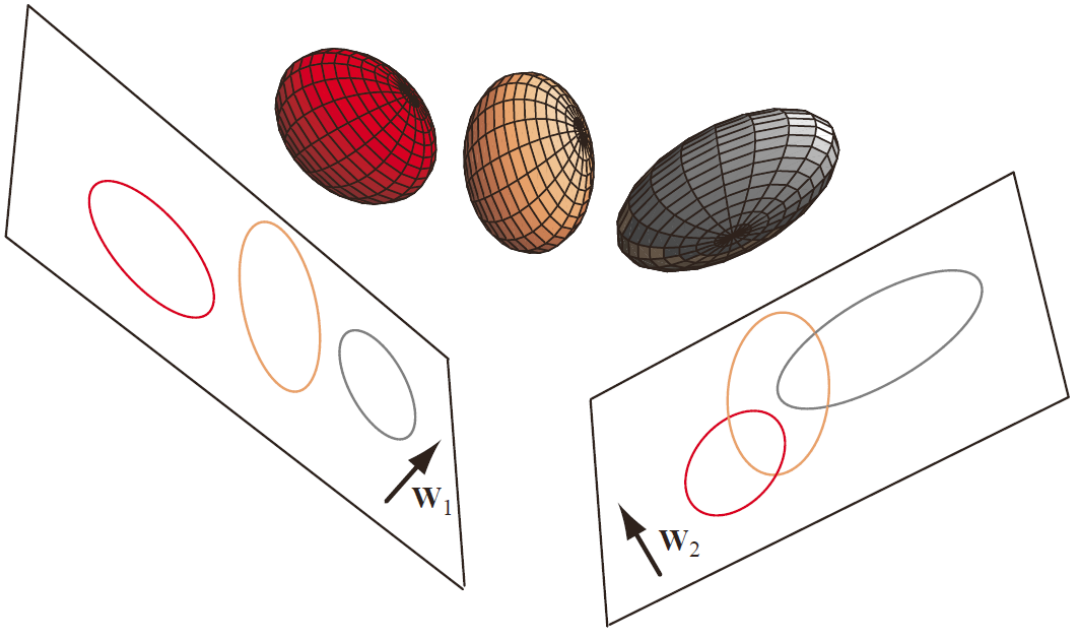


Figure 3.6: Three three-dimensional distributions are projected onto two-dimensional subspaces, described by a normal vectors $\mathbf{w_1}$ and $\mathbf{w_2}$. Informally, multiple discriminant methods seek the optimum such subspace, i.e., the one with the greatest separation of the projected distributions for a given total within-scatter matrix, here as associated with $\mathbf{w_1}$.

- These equations show how the within-class and between-class scatter matrices are transformed by the projection to the lower dimensional space (Fig. 3.6).

Multiple Discriminant Analysis

- We seek is a transformation matrix **W** that maximizes the ratio of the between-class scatter to the within-class scatter.
- A simple scalar measure of scatter is the determinant of the scatter matrix. The determinant is the product of the eigenvalues, and hence is the product of the "variances" in the principal directions, thereby measuring the square of the hyperellipsoidal scattering volume.
- Using this measure, we obtain the criterion function

$$J(\mathbf{W}) = \frac{|\tilde{\mathbf{S}}_B|}{|\tilde{\mathbf{S}}_W|} = \frac{|\mathbf{W}^t \mathbf{S}_B \mathbf{W}|}{|\mathbf{W}^t \mathbf{S}_W \mathbf{W}|}.$$

Multiple Discriminant Analysis

The problem of finding a rectangular matrix **W** that maximizes $J(\cdot)$ is tricky, though fortunately it turns out that the solution is relatively simple. The columns of an optimal **W** are the generalized eigenvectors that correspond to the largest eigenvalues in

$$\mathbf{S}_B \mathbf{w}_i = \lambda_i \mathbf{S}_W \mathbf{w}_i.$$

A few observations about this solution are in order. First, if $\mathbf{S_W}$ is non-singular, this can be converted to a conventional eigenvalue problem as before. However, this is actually undesirable, since it requires an unnecessary computation of the inverse of $\mathbf{S_W}$. Instead, one can find the eigenvalues as the roots of the characteristic polynomial

$$\left|\mathbf{S}_B - \lambda_i \mathbf{S}_W\right| = 0$$

and then solve

$$(\mathbf{S}_B - \lambda_i \mathbf{S}_W)\mathbf{w}_i = 0$$

directly for the eigenvectors $\mathbf{w_i}$.

Multiple Discriminant Analysis

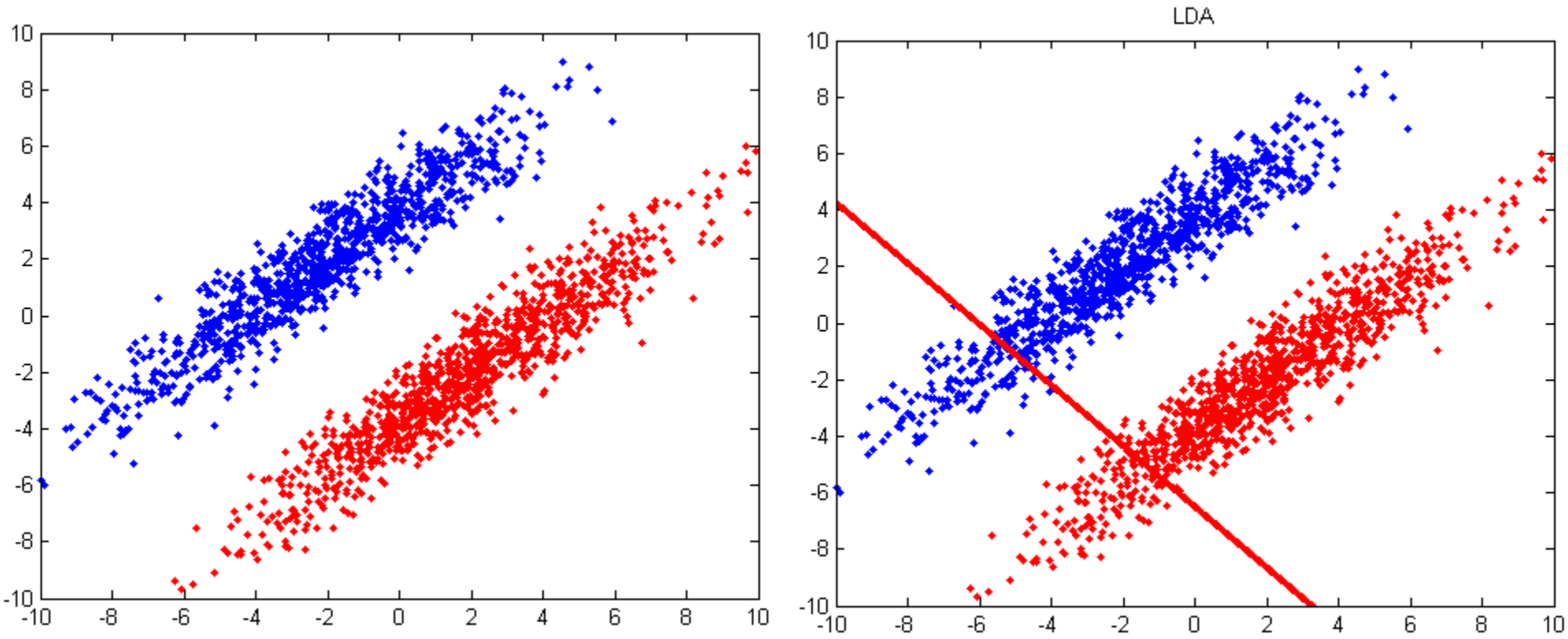
- Because $\mathbf{S_B}$ is the sum of c matrices of rank one or less, and because only $c-1$ of these are independent, $\mathbf{S_B}$ is of rank $c-1$ or less. Thus, no more than $c-1$ of the eigenvalues are nonzero, and so the new dimensionality is up to (c-1).

Multiple Discriminant Analysis

- Once we have projected the data onto the optimal subspace (defined as above), we can use the methods of Chapter 2 to create our full classifier.
- As in the two-class case, multiple discriminant analysis primarily provides a reasonable way of reducing the dimensionality of the problem. Parametric or nonparametric techniques that might not have been feasible in the original space may work well in the lower-dimensional space. In particular, it may be possible to estimate separate covariance matrices for each class and use the general multivariate normal assumption after the transformation where this could not be done with the original data.
- In general, if the transformation causes some unnecessary overlapping of the data and increases the theoretically achievable error rate, then the problem of classifying the data still remains.

LDA

$$Y_1 = \mathbf{w}^T \mathbf{X}_1 \qquad Y_2 = \mathbf{w}^T \mathbf{X}_2$$



Find a transformation **w**, such that the $\mathbf{w}^T\mathbf{X}_1$ and $\mathbf{w}^T\mathbf{X}_2$ are maximally separated & each class is minimally dispersed (maximum separation)

PCA vs. LDA

	PCA	LDA	
Sample mean	$\boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x_i}$	$\boldsymbol{\mu_i} = \frac{1}{N_i}\sum_{\mathbf{x}\in\omega_i} \mathbf{x}$	Mean for each class
Scatter matrix	$\mathbf{S} = \sum_{i=1}^{n}(x_i - \mu)(x_i - \mu)^T$	$\mathbf{S_w} = \sum_{i=1}^{c}\sum_{x_j\in C_i}(x_j - \mu_i)(x_j - \mu_i)^T$	Within-class scatter
		$\mathbf{S_b} = \sum_{i=1}^{c} N_i(\mu_i - \mu)(\mu_i - \mu)^T$	Between-class scatter
Eigen decomposition	$\mathbf{Sw} = \lambda\mathbf{w}$	$\mathbf{S_W^{-1}S_B w} = \lambda\mathbf{w}$	Eigen decomposition

$$\mathbf{Y} = \mathbf{w}^T\mathbf{X}$$

X is transformed to Y using w

Principal Component Analysis (PCA)

- Example
  - X={(4,1),(2,4),(2,3),(3,6),(4,4)}
  - Statistics

$$\boldsymbol{\mu} = [3.0 \quad 3.6],$$
$$\mathbf{S} = \begin{bmatrix} 4.0 & -2.0 \\ -2.0 & 13.2 \end{bmatrix}$$

  - Solve the Eigen value problem

$$\mathbf{S}\mathbf{w} = \lambda\mathbf{w}$$

Linear Discriminant Analysis (LDA)

- Example
  - $\mathbf{X}_1$= {(4,1),(2,4),(2,3),(3,6),(4,4)}
  - $\mathbf{X}_2$= {(9,10),(6,8),(9,5),(8,7),(10,8)}
  - Class statistics

$$\boldsymbol{\mu}_1 = [3.0 \quad 3.6], \quad \boldsymbol{\mu}_2 = [7.67 \quad 7.0], \quad \boldsymbol{\mu} = [5.7 \quad 5.6]$$

$$\mathbf{S}_1 = \begin{bmatrix} 4.0 & -2.0 \\ -2.0 & 13.2 \end{bmatrix}, \quad \mathbf{S}_2 = \begin{bmatrix} 11.89 & 2.0 \\ 2.0 & 15.0 \end{bmatrix}$$

  - Within and between class scatter

$$\mathbf{S}_B = \begin{bmatrix} 72 & 54 \\ 54 & 40 \end{bmatrix}, \quad \mathbf{S}_W = \begin{bmatrix} 15.89 & 0.0 \\ 0.0 & 28.2 \end{bmatrix}$$

  - Solve the Eigen value problem

$$\mathbf{S}_W^{-1}\mathbf{S}_B\mathbf{w} = \lambda\mathbf{w}$$